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Solution

Dimensional formula for

$$\frac{1}{2}mv^2 = [M][LT^{-1}]^2 = [ML^2T^{-2}]$$

Dimensional formula for

$$mgh = [M][LT^{-2}][L] = [ML^2T^{-2}]$$

$$[ML^2T^{-2}] = [ML^2T^{-2}]$$

Both sides are dimensionally the same, hence the equations $\frac{1}{2}mv^2 = mgh$ is dimensionally correct.

(iii) To establish the relation among various physical quantities

If the physical quantity Q depends upon the quantities Q_1, Q_2 and Q_3 ie. Q is proportional to Q_1, Q_2 and Q_3.

Then,

$$Q \alpha Q_1^{\ a} Q_2^{\ b} Q_3^{\ c}$$
$$Q = k\, Q_1^{\ a} Q_2^{\ b} Q_3^{\ c}$$

where k is a dimensionless constant. When the dimensional formula of Q, Q_1, Q_2 and Q_3 are substituted, then according to the principle of homogeneity, the powers of M, L, T are made equal on both sides of the equation. From this, we get the values of a, b, c

EXAMPLE 1.15

Obtain an expression for the time period T of a simple pendulum. The time period T depends on (i) mass '*m*' of the bob (ii) length '*l*' of the pendulum and (iii) acceleration due to gravity *g* at the place where the pendulum is suspended. (Constant k = 2π) i.e

Solution

$$T \alpha m^a l^b g^c$$
$$T = k.\quad m^a l^b g^c$$

Here k is the dimensionless constant. Rewriting the above equation with dimensions

$$[T^1] = [M^a]\,[L^b]\,[LT^{-2}]^c$$
$$[M^0L^0T^1] = [M^a\, L^{b+c}\, T^{-2c}]$$

Comparing the powers of M, L and T on both sides, a=0, b+c=0, -2c=1

Solving for a,b and c a = 0, b = 1/2, and c = −1/2

From the above equation T = k. m^0 $\ell^{1/2}$ $g^{-1/2}$

$$T = k\left(\frac{\ell}{g}\right)^{1/2} = k\sqrt{\ell/g}$$

Experimentally k = 2π, hence $T = 2\pi\sqrt{\ell/g}$

EXAMPLE 1.16

The force F acting on a body moving in a circular path depends on mass of the body (m), velocity (v) and radius (r) of the circular path. Obtain the expression for the force by dimensional analysis method. (Take the value of k=1)

Solution

$$F \alpha m^a v^b r^c$$
$$F = k\, m^a v^b r^c$$